each other very well. As the mathematician Hans Freudenthal put it, "Lie and Klein had quite different characters as humans and mathematicians: the algebraist Klein was fascinated by the peculiarities of charming problems; the analyst Lie, parting from special cases, sought to understand a problem in its appropriate generalization" [**16**, p. 323].

Lie liked to bounce ideas off his friend's head, and Klein's returns were often quite powerful. In particular, Klein pointed out an analogy between Lie's research on the tetrahedral complex and the Galois theory of commutative permutation groups. Blissfully unaware of the difficulties on his path, Lie enthusiastically embraced this suggestion. Developing a continuous analog of the Galois theory of algebraic equations became Lie's *idée fixe* for the next several years.

Lie and Klein traveled to Paris together, and there Lie produced the famous contact transformation, which mapped straight lines into spheres. An application of this expertise to the Earth sphere, however, did not serve him well. After the outbreak of the Franco-Prussian war, Lie could not find a better way to return to Norway than by first hiking to Italy. With his peculiar hiking habits, such as taking off his clothes in the rain and putting them into his backpack, he was not able to flee very far. The French quickly apprehended him and found papers filled with mysterious symbols. Lie's efforts to explain the meaning of his mathematical notation did not dispel the authorities' suspicion that he was a German spy. A short stay in prison afforded him some quiet time to complete his studies, and upon return to Norway, Lie successfully defended his doctoral dissertation. Unable to find a job in Norway, Lie resolved to go to Sweden, but Norwegian patriots intervened, and the Norwegian National Assembly voted by a large majority to establish a personal extraordinary professorship for Lie at the University of Christiania. Although the salary offered was less than extraordinary, he stayed.

Lie's research on sphere mapping and his lively exchanges with Klein led both of them to think of more general connections between group theory and geometry. In 1872 Klein presented his famous Erlangen Program, in which he suggested unifying specific geometries under a general framework of projective geometry and using group